

3(a)	polar / water soluble / hydrophilic, substances cannot pass through the, phospholipid bilayer / hydrophobic core (in cell surface membranes) ; I too large / active transport / facilitated diffusion / (hydrophobic) fatty acid 'tails'	1
3(b)(i)	rate of uptake, becomes constant / reaches a plateau or at <u>high</u> concentrations / concentrations of $> 150 \mu\text{mol dm}^{-3}$, rate of uptake remains constant ; A levels off protein carriers / (hexose) transporters / VvHT1s, are working at their highest rate / are saturated ; A the number of protein carriers is limiting factor / AW A V_{max} is reached <i>in context of transporters</i>	2

Question	Answer	Marks
3(b)(ii)	any two from: 1 size / quality of grapes / grapevines 1 grapes will, be sweeter / have improved taste ; 2 idea that concentration of sugars in grapes determines uptake of water by osmosis ; 3 idea of more energy for growth of the grapes ; 4 larger yield ; 5 increased, profit / income, for farmers ; 6 AVP ; e.g. producing grapes in shorter time	2
3(c)(i)	(foreign) protein / glycoprotein, that stimulates, an immune response / production of antibodies / activation of lymphocytes ; A polysaccharide / molecule / foreign substance / foreign antigen	1

Question	Answer	Marks
3(c)(ii)	any four from: accept <i>T-cells</i> for <i>T-lymphocytes</i> 1 (group / clone of) T-lymphocytes have receptors <u>complementary</u> to an antigen ; A <u>specific</u> T-lymphocytes have receptors to an antigen 2 (specific) T-lymphocytes, bind / AW, to antigen (on surface of antigen-presenting cell / macrophage) ; 3 (selected) T-lymphocytes, divide many times (by mitosis) / undergo clonal expansion ; 4 T-helper cells secrete, cytokines / interleukins / interferon ; 5 detail of effect of cytokine ; e.g. activation of B-lymphocytes / increased phagocytosis by macrophages / 'angry' macrophages 6 T-killer cells, bind / attach, to infected cells and destroy them ; 7 detail of action of T-killer cells ; e.g. <i>ref. to</i>, perforin / granzyme / hydrogen peroxide / toxins 8 production of memory cells (in context of T-lymphocytes / T-helper cells / T-killer cells) ;	4

Question	Answer	Marks
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